

PRIYANSHU SHEKHAR

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EDUCATION

University of Southern California (USC)

June 2024 – May 2026

Master of Science in Computer Science (Specialization: Artificial Intelligence)

Los Angeles, California

Relevant Coursework: Analysis of Algorithms, Foundations of AI, Machine Learning Systems, Deep Learning and Its Applications, Applied Natural Language Processing, Robotics, Multi-modal Probabilistic Learning of Human Communication, Web Technologies

California State Polytechnic University-Pomona (Cal Poly Pomona)

August 2021 – May 2024

Bachelor of Science in Computer Science (Minor: Data Science) | Summa Cum Laude

Pomona, California

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Systems, Distributed Systems, Probability and Statistics, Big Data Analytics & Cloud Computing, Web Search & Recommender Systems, Systems Programming

PROFESSIONAL EXPERIENCE

Research Assistant

October 2025 – May 2026

University of Southern California

Los Angeles, California

- Designed and implemented Python-based data processing and validation pipelines for LLM security evaluation, translating research requirements into multi-step testing workflows and systematically validating outputs across complex privacy attack scenarios
- Designed graph-based synthetic datasets spanning 3+ sensitive domains (financial, medical, and personal), supporting retrieval-driven evaluation of information extraction attacks and responsible AI validation
- Translated methodologies from 25+ technical publications into executable testing procedures, documenting evaluation logic, comparing alternative approaches, and using experimental results to guide system architecture and implementation decisions
- Collaborated with researchers throughout design, implementation, testing, and iteration, converting evolving technical requirements into prototype functionality and refining backend workflows based on evaluation results

AI Engineer Intern

February 2024 – May 2024

Google

Pomona, California

- Engineered Python backend services, REST interfaces, and a centralized MySQL data layer supporting analysis of 50,000+ records, translating research requirements into maintainable application and database system
- Designed SQL queries and Pandas processing workflows for data extraction, transformation, validation, and reporting, improving reporting accuracy and enabling researchers to analyze large experimental datasets consistently
- Developed a full-stack React and JavaScript application integrated with backend APIs, creating modular software components that enabled non-technical stakeholders to inspect data and interact with analytical workflows
- Collaborated with cross-functional teams throughout the software development lifecycle, balancing implementation trade-offs across design, testing, deployment, and continuous improvement

Machine Learning Engineer

May 2023 – February 2024

California State Polytechnic University-Pomona

Pomona, California

- Re-engineered an existing STARFM ML-based forecasting implementation into USTARFM in Python, identifying limitations in the original workflow and implementing algorithmic and data-processing changes that improved prediction accuracy by 15%
- Designed a distributed AWS EC2 processing pipeline for 300+ satellite images, configuring workloads across edge and cloud infrastructure to support repeatable processing, model execution, and system validation
- Built feature engineering and preprocessing pipelines for large-scale geospatial datasets, improving model performance by 25%
- Analyzed system bottlenecks and evaluated alternative workload configurations, partitioning processing between edge devices and cloud infrastructure to reduce latency by 35% and improve operational scalability

PERSONAL PROJECTS

ASLTutor – Multimodal ASL Learning Platform

January 2026 – May 2026

- Built an AI-native learning platform using FastAPI, React, Whisper ASR, and LLM that delivered real-time multimodal ASL tutoring via backend APIs
- Designed a computer vision workflow combining MediaPipe and Graph Neural Networks to generate adaptive feedback through AI inference
- Optimized backend APIs and inference orchestration to improve response latency while maintaining production-ready software architecture

Event Finder Web and Android Application

September 2025 – January 2026

- Built and deployed a full-stack application using React, TypeScript/JavaScript, Node.js, REST APIs, MongoDB, and Google Cloud, integrating multiple external systems into a unified event-discovery workflow
- Designed MongoDB data models, aggregation queries, and backend interfaces supporting reliable data exchange across modular services
- Integrated Ticketmaster, Spotify, Google Maps APIs, testing interface behavior and normalizing data from external systems into consistent workflows

Text-to-SQL Application

January 2025 – April 2025

- Engineered a Text-to-SQL system using DeepSeek-Coder, LoRA, and RAG, translating natural-language requirements into schema-aware SQL queries
- Built modular Python services using Hugging Face, Sentence Transformers, SQLite, and REST APIs, simplifying testing, debugging, and configuration
- Implemented SQL validation, metadata management, and inference optimizations, improving database-query reliability and application responsiveness

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, Python, SQL, C Bash

Web Development: React, Node.js, Flask, Django, FastAPI, RESTful APIs, TailwindCSS

Data Management: MySQL, MongoDB, SQLite, Hadoop

AI & ML Frameworks: Prompt Engineering, PyTorch, TensorFlow, Scikit-learn, Hugging Face, RAG, AI Agents, NumPy, Pandas

Cloud & Infrastructure: AWS (EC2), Google Cloud Platform, Docker, Kubernetes, Git, Microservices Architecture